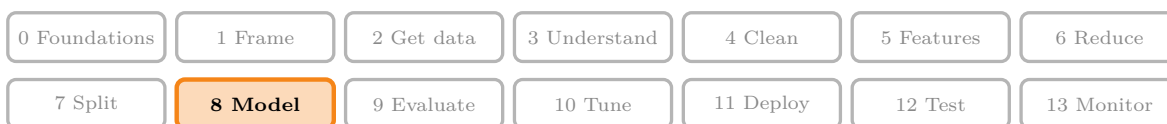


Introduction to Random Forest

Contents

1. Overview	2
2. Why random forests are so popular	3
3. Where the name comes from	3
4. How a random forest works	4
4.1 Building one tree	4
4.2 Using the forest	5
4.3 Checking the forest	6
5. A random forest by hand	6
5.1 The data and the sampling functions	7
5.2 Row sampling	8
5.3 Column sampling and combined sampling	8
5.4 What a real random forest does differently	9
6. Summary	11
7. Sources	11
8. Key terms	11

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Overfitting ([Note 91](#)); Hyperparameter tuning ([Note 98](#)); Decision trees ([Note 100](#)); Feature importance ([Note 100](#)); Bagging ([Note 107](#)); OOB score ([Note 107](#)).
- **Leads to:** Balanced random forest ([Note 133](#)).
- **Compare with:** Bagging ([Note 107](#)); Dropout ([Note 1024](#)).

1. Overview

Key point: A random forest trains many decision trees, each on its own random sample of the data and with a random choice of features at every split, and combines their answers by majority vote (classification) or mean (regression).

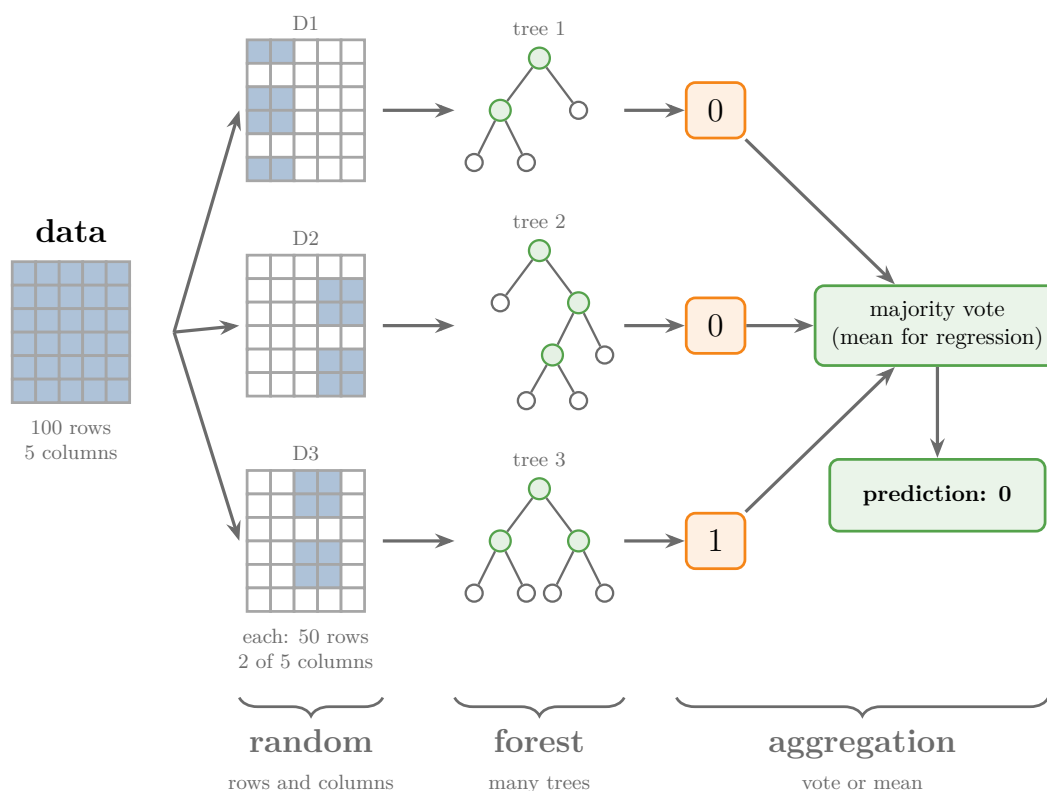


Figure 1: A random forest in outline, drawn for the three-tree forest built by hand in section 5: each tree gets its own random data, grows differently, and votes

A **random forest** (G-1611) is **bagging** (G-251; the [bagging Note](#)) with **decision trees** (G-561; the [decision trees Note](#)) as the **base models** (G-260), the models the ensemble

combines. Figure 1 shows the whole idea: random data goes to different trees, each tree predicts, and the forest takes the majority.

This Note covers:

- why random forests are so widely used (section 2);
- where the name comes from (section 3);
- how a random forest is built, used and checked (section 4);
- a small forest of three trees built by hand (section 5).

The next Notes explain:

- why it works so well ([random forest and bias-variance](#));
- how it differs from plain bagging ([bagging vs random forest](#));
- its [hyperparameters](#) and their [tuning](#);
- the [OOB score](#) and [feature importance](#).

The Notebook (`notebook.ipynb`) runs the forest built by hand.

2. Why random forests are so popular

Key point: Strong results on many problems, classification and regression alike, with little tuning.

Three things make the random forest one of the first algorithms to try in a project:

1. **Strong results on many problems.** On many problems a random forest performs about as well as boosting (ESL §15.1), a family of methods (**boosting**, G-318) covered in later Notes. In a comparison of 179 classifiers on 121 datasets, random forests were the best family overall (Fernández-Delgado et al., 2014).
2. **Both problem types.** `RandomForestClassifier` predicts classes and `RandomForestRegressor` predicts numbers.
3. **Little tuning.** Random forests are simpler to train and tune than boosting (ESL §15.1), and the default settings usually give a good result. Little tuning makes the random forest especially friendly for beginners.

Extra: The random forest was published by the statistician Leo Breiman in 2001, in a paper titled *Random Forests* (Breiman, 2001). Breiman had also introduced bagging (Breiman, 1996). The name and the idea of training trees on random subsets of the features came earlier, from Tin Kam Ho’s “random decision forests” (Ho, 1995).

3. Where the name comes from

Key point: “Forest” because it is a collection of trees; “random” because each tree gets randomly sampled data.

- **Forest:** we train many decision trees together. A group of trees is a forest.

- **Random:** a random forest is a bagging technique, and bagging means **bootstrap aggregation**. Bootstrapping draws a random sample of the data for each tree (the [bagging Note](#), section 2.1). That random selection gives the “random”.

So a random forest is a special case of bagging: bagging lets the base model be any algorithm, and when that algorithm is a decision tree, we call the result a random forest. There is one more, subtler difference, the subject of the [bagging vs random forest Note](#).

4. How a random forest works

Key point: For each tree: draw a bootstrap sample of the observations, then grow a tree that looks at only a few randomly chosen features at every split. To predict, send the new point to every tree and take the vote or the mean.

Here an **observation** is one record (a row of the data table), a **feature** is one input variable (a column), and the **target** is the output we predict.

4.1 Building one tree

Key point: Two sources of randomness: which observations the tree sees, and which features each split may use.

Think of a panel of doctors. Each doctor has seen a different set of past patients, and for each question is allowed to look at only a couple of test results picked at random. The doctors end up reasoning in different ways, and their mistakes differ, so the panel’s majority is better than any one doctor.

Figure 2 builds one tree this way on 12 patients of the heart disease data (the data of the [tuning Note](#)) with 5 features:

1. **Draw a bootstrap sample.** Pick 12 patients at random **with replacement**: the same patient can be picked again. The result is a **bootstrap sample** (G-319), as large as the data. In Figure 2, patient 1 is drawn three times, and patients 2, 5, 6, 9 and 12 are never drawn.
2. **Grow a tree, with a random choice of features at every split.** At the first split the tree does not compare all 5 features. Two are drawn at random, pressure and cholesterol, and the better of the two (cholesterol) makes the split. At the next split two features are drawn again, and so on until every leaf is pure.
3. **Repeat** steps 1 and 2 for every tree, usually a hundred or more.

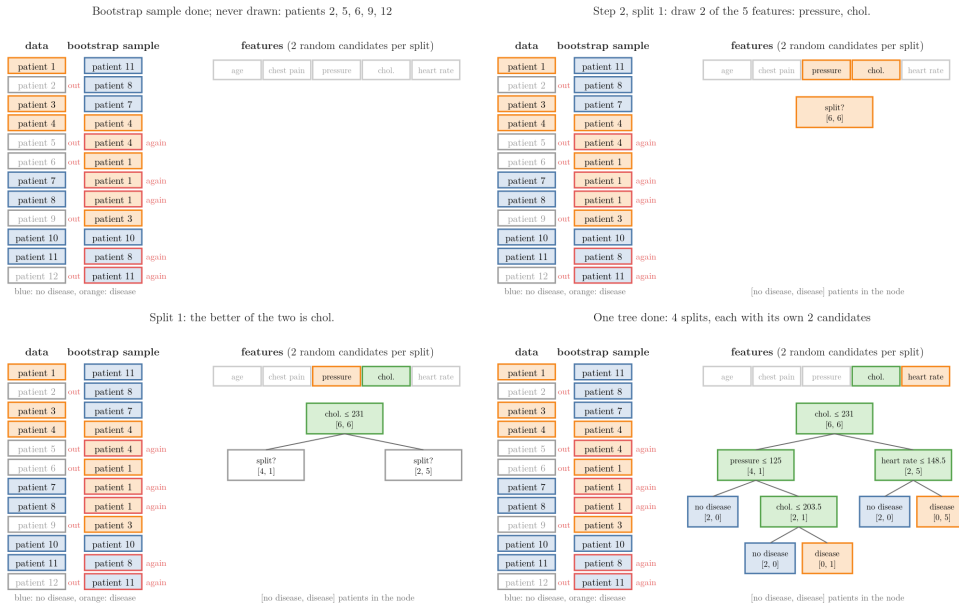


Figure 2: One tree of a random forest, built on 12 patients of the heart disease data. Step 1: a bootstrap sample (“again”: drawn before; “out”: never drawn). Step 2: at every split, 2 of the 5 features are drawn at random (orange) and the better one (green) makes the split. Idea after StatQuest, “Random Forests Part 1”

Watch the feature row at the top of Figure 2: the two orange candidates change from split to split.

Drawing the features again at every split is **node-level column sampling** (G-1325). The number of features drawn is the setting `max_features` (G-1185). scikit-learn’s default for classification is the square root of the number of features: with 5 features, $\sqrt{5} = 2.2$, rounded down to **2**, as in Figure 2.

Because every tree sees different observations and different features at each split, every tree learns a different structure. The variety of the trees is what makes the forest better than a single tree (the [random forest and bias-variance Note](#)).

4.2 Using the forest

Key point: Every tree predicts; the forest returns the majority vote or the mean.

A new **query point** (G-1605), the observation we want a prediction for, goes to every tree. The combining step is **aggregation** (G-183): the forest returns the **majority vote** (G-1146) for classification or the mean for regression, as every ensemble does (the [ensemble learning Note](#), section 3). For example, if 57 of 100 trees say 1 and 43 say 0, the forest predicts 1.

Figure 3 builds a forest of 100 trees on the student placement data of the [toy project Note](#) (two features, CGPA and IQ, 100 students). On the left, each new tree’s bootstrap sample: bigger dots were drawn more than once, grey crosses were left out (about a third). The **decision boundary** (G-555) is the line where the predicted class switches from “not placed”

to “placed”. Watch each single tree put narrow strips of the wrong class around a few points, while the forest’s vote map on the right settles into one clean decision boundary near CGPA 6; after about 20 trees, adding more changes it very little.

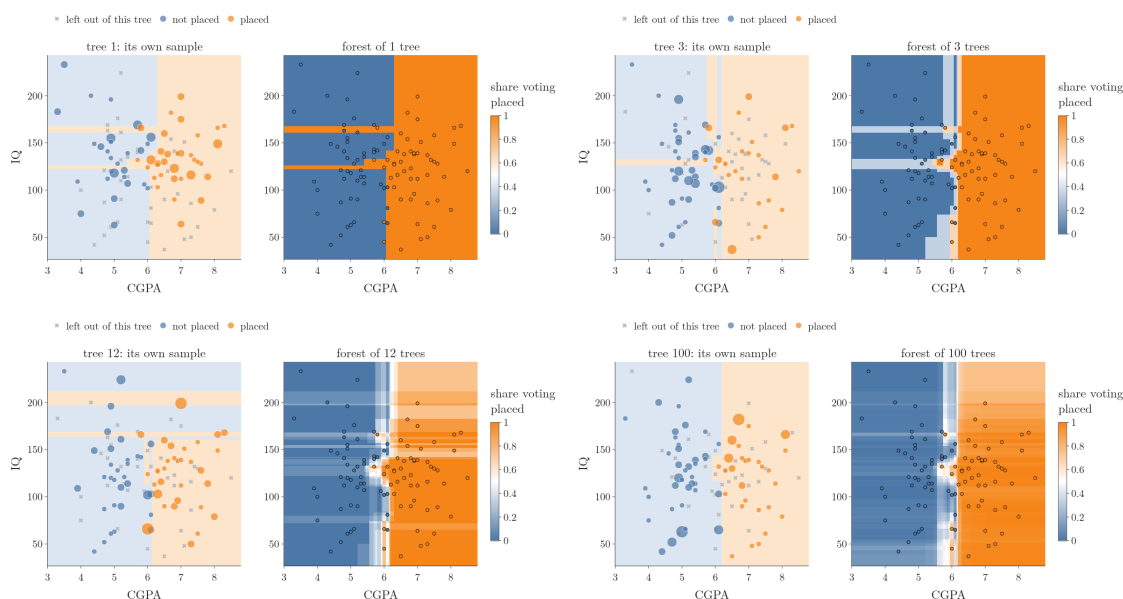


Figure 3: A random forest grows. Left: the newest tree’s own bootstrap sample and its regions. Right: the share of all trees so far that vote “placed”. Each tree is a bootstrap sample plus a fully grown tree with a random feature choice at each split, as in `RandomForestClassifier`. Growing the forest tree by tree follows StatQuest’s “Random Forests Part 1” (Starmar)

4.3 Checking the forest

Key point: The observations a tree never drew can test that tree, so a forest can be checked without a separate test set.

Each tree leaves out about a third of the observations (the patients marked “out” in Figure 2). The left-out observations of a tree are its **out-of-bag** observations. Each observation is predicted by only the trees that never saw it, and the share predicted correctly is the **out-of-bag score** (G-1411): the [OOB score Note](#) explains it in full.

5. A random forest by hand

Key point: Three trees, each trained on its own random subset of a dataset of 100 observations, vote on one query point; with row, column or combined sampling, the majority gets it right even when one tree is wrong.

To see every step, we build a forest of three trees ourselves (Figure 1). Our forest is a simplified version: it has only three trees, and it picks each tree’s features **once per tree** instead of at every split (section 5.4).

Each tree gets its own random subset of the data, made in one of three ways:

- **row sampling** (G-1712): random observations, all features;
- **column sampling**, also called **feature sampling** (G-413): all observations, random features;
- **combined sampling** (G-418): random observations and random features.

These three ways are the bagging, random subspaces and random patches types of the [bagging Note](#), section 6.

5.1 The data and the sampling functions

Key point: 100 observations, 5 features, two classes; three small functions draw the subsets.

We make a classification dataset of 100 observations with `make_classification` (the [perceptron code Note](#)): 5 features, in columns `col1` to `col5`, and a target in the column `target`, with classes 0 and 1.

Python: The three ways of sampling.

```
def sample_rows(df, share):
    n = int(share * len(df))
    return df.sample(n, replace=True)

def sample_features(df, share):
    # do not count the target column
    n = int(share * (df.shape[1] - 1))
    cols = list(rng.choice(df.columns[:-1], n,
                           replace=False))
    return df[cols + ["target"]]

def combined_sampling(df, row_share, col_share):
    rows = sample_rows(df, row_share)
    return sample_features(rows, col_share)
```

`df.sample(n, replace=True)` draws `n` rows with replacement (the [bagging Note](#), section 5.2). `rng.choice(columns, n, replace=False)` picks `n` different column names at random; `rng` is NumPy's random generator, `np.random.default_rng(4)`, seeded so every run draws the same subsets. The target column is added back, because every tree needs it to learn.

Each tree is a fully grown `DecisionTreeClassifier`, trained on its subset's features. The query point is observation 7 of the data, whose true class is **0**.

5.2 Row sampling

Key point: 20 observations per tree, with replacement: the votes are 1, 0, 0, so the forest says 0.

With `sample_rows(df, 0.2)`, each tree gets 20 observations (20% of 100), drawn with replacement:

Tree	Observations	Repeated	Depth	Vote
1	20	2	2	1
2	20	1	2	0
3	20	1	2	0

Tree 1 is wrong, but two trees say 0, so the majority, **0**, is right. Each tree saw different observations, so each placed its decision boundaries in different places (Figure 4, top row).

5.3 Column sampling and combined sampling

Key point: 4 of 5 features per tree: all three trees say 0. Half the observations and 2 of 5 features: the votes are 0, 0, 1, so the forest still says 0.

With `sample_features(df, 0.8)` (Figure 4, middle row), each tree gets all 100 observations and 4 of the 5 features (80% of 5). The three trees got different feature sets: (col5, col4, col3, col2), (col5, col2, col1, col3) and (col1, col3, col2, col5). With all the observations, the trees grow deeper (depths 6, 3 and 3), and all three vote **0**.

With `combined_sampling(df, 0.5, 0.5)` (Figure 4, bottom row), each tree gets 50 observations (10 to 14 of them repeats) and 2 of the 5 features (half of 5 is 2.5, and `int` rounds down to 2). Figure 1 shows this setting:

Tree	Features	Depth	Vote
1	col2, col1	4	0
2	col5, col4	5	0
3	col4, col3	6	1

Combined sampling adds the most randomness: the trees differ in both observations and features. The majority is again **0**, the right answer.

	three trees, each on its own random subset			the forest
row sampling 20 obs., all 5 features	tree 1 votes 1 all 5 features depth 2	tree 2 votes 0 all 5 features depth 2	tree 3 votes 0 all 5 features depth 2	majority: 0 2 of 3 votes
column sampling 100 obs., 4 features	tree 1 votes 0 col5, col4, col3, col2 depth 6	tree 2 votes 0 col5, col2, col1, col3 depth 3	tree 3 votes 0 col1, col3, col2, col5 depth 3	majority: 0 3 of 3 votes
combined 50 obs., 2 features	tree 1 votes 0 col2, col1 depth 4	tree 2 votes 0 col5, col4 depth 5	tree 3 votes 1 col4, col3 depth 6	majority: 0 2 of 3 votes

Figure 4: The forest built by hand: for each sampling setting, the three trees' votes on the query point (true class 0), with each tree's features and depth; a black frame marks a wrong vote. The right column is the forest's majority

Figure 4 puts the three settings side by side. Watch the black-framed boxes: one tree is wrong in two of the settings, and the majority on the right is still the true class 0 every time.

Python: Asking one tree about the query point.

```
inputs = d.columns[:-1] # this tree's input columns
tree = DecisionTreeClassifier().fit(d[inputs], d["target"])
tree.predict(query[inputs])
```

A tree trained on 2 features must be given exactly those 2 columns when it predicts. Selecting `query[inputs]` passes the same columns in the same order.

5.4 What a real random forest does differently

Key point: scikit-learn's `RandomForestClassifier` does all this in one line, with 100 trees by default, and samples the features afresh at every split.

Python: A random forest in scikit-learn.

```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(random_state=42)
rf.fit(df.iloc[:, :5], df["target"])
rf.predict(query.iloc[:, :5]) # array([0])
```

The forest of 100 trees also predicts 0.

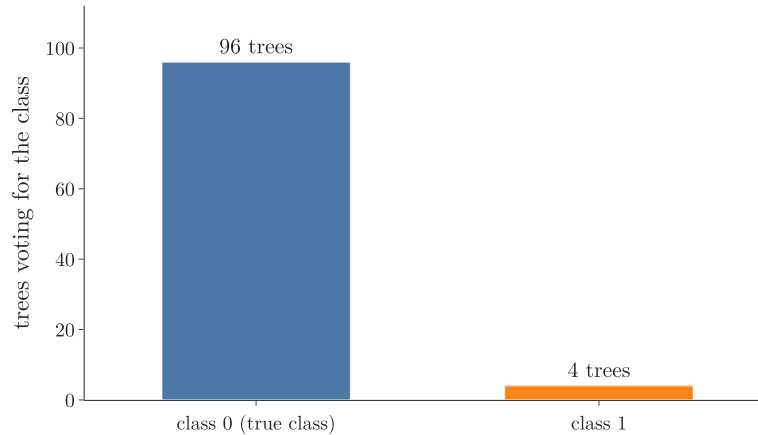


Figure 5: How the 100 trees of RandomForestClassifier vote on the same query point (true class 0)

Figure 5 counts the votes: 96 trees say 0 and 4 say 1. With 100 voters, a few wrong trees barely dent the majority.

Two things differ from our hand-built version:

- **More trees.** Three trees are only enough to show the idea; the default is 100.
- **Feature sampling at every split.** Our functions picked the features once per tree. A random forest picks a fresh random set of features at every node of every tree (section 4.1; Breiman, 2001, section 4), which makes the trees even less alike (ESL §15.2). Sampling at every split is the main difference between bagging and a random forest, explained in the [bagging vs random forest Note](#).

6. Summary

Step	What happens	In our example
Bootstrapping	each tree gets a bootstrap sample of the observations	by hand: 3 trees; 20 observations, or 4 features, or 50 observations and 2 features each
Training	each tree is fully grown, with a few random features to choose from at every split	trees of depth 2 to 6, all different
Aggregation	majority vote (classes) or mean (numbers)	votes 1, 0, 0 and 0, 0, 1: prediction 0

- A random forest is bagging with decision trees as the base models.
- “Forest”: many trees. “Random”: each tree gets randomly sampled data.
- Each tree is built from a bootstrap sample, and each split chooses among a few randomly drawn features (`max_features`, by default the square root of the number of features).
- The observations a tree never drew are its out-of-bag observations; they give a free check of the forest.
- A random forest works for classification and regression, and gives strong results with little tuning (ESL §15.1).

7. Sources

Built from

- CampusX, “Introduction to Random Forest | Intuition behind the Algorithm”, YouTube, <https://www.youtube.com/watch?v=F9uESCHGjhA>
- StatQuest with Josh Starmer, “StatQuest: Random Forests Part 1 - Building, Using and Evaluating”, YouTube, https://www.youtube.com/watch?v=J4Wdy0Wc_xQ (the two building steps and Figures 2 and 3)

Other references

- Breiman, L. (1996). “Bagging Predictors”. *Machine Learning* 24(2), 123–140.
- Breiman, L. (2001). “Random Forests”. *Machine Learning* 45(1), 5–32.
- Fernández-Delgado, M., Cernadas, E., Barro, S. and Amorim, D. (2014). “Do we Need Hundreds of Classifiers to Solve Real World Classification Problems?” *Journal of Machine Learning Research* 15, 3133–3181.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning* (ESL), 2nd ed. Springer, sections 15.1 and 15.2.
- Ho, T. K. (1995). “Random Decision Forests”. *Proceedings of the 3rd International Conference on Document Analysis and Recognition*, 278–282.

8. Key terms

Term	Meaning
Random forest	Bagging with decision trees as the base models; the trees vote (classification) or are averaged (regression)
Observation	One record of the data: one row of the data table
Feature	An input variable: one column of the data table
Target	The output we predict
Row sampling	Giving each base model a random subset of the observations
Column sampling (feature sampling)	Giving each base model a random subset of the features
Combined sampling	Giving each base model random observations and random features together
RandomForestClassifier	scikit-learn's random forest for classification
RandomForestRegressor	scikit-learn's random forest for regression